
Practice 9 – State Feedback Gain Matrix for Rotatory Inverted Pendulum Part A. State Space Modeling

Abstract

The rotatory inverted pendulum, known as Furuta pendulum, is a system used in control labs that provides a compact yet impressive platform for control demonstrations and draws the attention of the students to develop their own control laws. It was invented in 1992 at Tokyo Institute of Technology by Katsuhisa Furuta and consists of a driven arm which rotates in the horizontal plane and a pendulum attached to that arm which is free to rotate in the vertical plane. The pendulum is underactuated and extremely non-linear due to the gravitational forces and the coupling arising from the Coriolis and centripetal forces. The “swing-up” problem for the Furuta pendulum was studied using energy methods by Åström and Furuta. In this lab, you must obtain and validate the linear time invariant state space representation for the Pendulum DOWN / Crane Configuration.

Topics Covered

- First principles modeling
- State space representation
- Model validation

Prerequisites

- Hardware Interfacing laboratory experiment
- Rotary Pendulum Modeling laboratory experiment

1. Introduction

The rotary pendulum model is shown in Figure 1.1. The rotary arm pivot is attached to the Qube Servo 3 system and is actuated. The arm has a length of r , a moment of inertia of J_r , and its angle $\theta(t)$ increases positively when it rotates counterclockwise (CCW). The servo (and thus the arm) should turn in the CCW direction when the control voltage is positive, $v_m(t) > 0$.

The pendulum link is connected to the end of the rotary arm. It has a total length of L_p and its center of mass is at $l = L_p/2$. The moment of inertia about its center of mass is J_p . The rotary pendulum angle $\alpha(t)$ is zero when it is hanging downward and increases positively when rotated CCW.

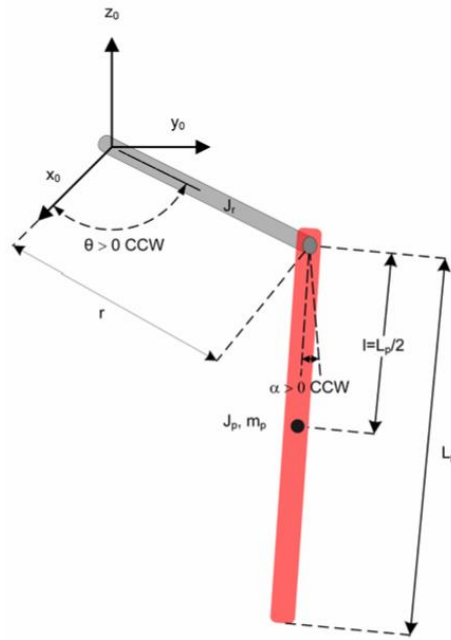


Fig. 1.1: Rotary pendulum model in DOWN / Crane Configuration.

The equations of motion (EOM) for the pendulum system were developed using the Euler Lagrange method. This systematic method is often used to model complicated systems such as robot manipulators with multiple joints. The total kinetic and potential energy of the system is obtained, then the Lagrangian can be found. A number of derivatives are then computed to yield the EOMs. The resultant nonlinear EOM are:

$$\begin{aligned} \{J_r + J_p \sin[\alpha^2(t)]\} \ddot{\theta}(t) + m_p l r \cos[\alpha(t)] \ddot{\alpha}(t) + \dots \\ + 2J_p \sin[\alpha(t)] \cos[\alpha(t)] \dot{\theta}(t) \dot{\alpha}(t) - m_p l r \sin[\alpha(t)] \dot{\alpha}^2(t) \end{aligned} = \tau - b_r \dot{\theta}(t) \quad (1.1)$$

and

$$\begin{aligned} J_p \ddot{\alpha}(t) + m_p l r \cos[\alpha(t)] \ddot{\theta}(t) - \dots \\ - J_p \sin[\alpha(t)] \cos[\alpha(t)] \dot{\theta}^2(t) + m_p g l \sin[\alpha(t)] \end{aligned} = -b_p \dot{\alpha}(t) \quad (1.2)$$

where:

- $J_r = m_r r^2 / 3$ is the moment of inertia of the rotary arm with respect to the pivot (i.e. rotary arm axis of rotation)
- $J_p = m_p L_p^2 / 3$ corresponds to the moment of inertia of the pendulum link relative to the pendulum pivot (i.e. axis of rotation of pendulum)
- b_r represents the viscous damping acting on the rotary arm
- b_p stands for the viscous damping acting on the pendulum link

Moreover, the applied torque at the base of the rotary arm generated by the servo motor is:

$$\tau = \frac{k_m}{R_m} [v_m(t) - k_m \dot{\theta}(t)] \quad (1.3)$$

These equations of motion are based on the Furuta Pendulum model in the Pendulum Equations document by Dr. K. J. Åström, located in the System Models folder. Note that the equations in that document are for the rotary inverted pendulum, i.e. the equation above are for the pendulum hanging downwards.

In order to use linear techniques to, for instance, assess the stability of the system or design a linear feedback controller, we can approximate the system by linearizing the original nonlinear equations of motions about the zero angle for both the rotary arm and the pendulum, i.e. $\theta(t) = 0$ and $\alpha(t) = 0$; then, the resultant linear EOM for the rotary pendulum are defined as:

$$J_r \ddot{\theta}(t) + m_p l r \ddot{\alpha}(t) = \tau - b_r \dot{\theta}(t) \quad (1.4)$$

and

$$J_p \ddot{\alpha}(t) + m_p l r \ddot{\theta}(t) + m_p g l \alpha(t) = -b_p \dot{\alpha}(t) \quad (1.5)$$

Solving for the acceleration terms yields:

$$\ddot{\theta}(t) = \frac{1}{J_t} [m_p^2 l^2 r g \alpha(t) - J_p b_r \dot{\theta}(t) + m_p l r b_p \dot{\alpha}(t) + J_p \tau] \quad (1.6)$$

and

$$\ddot{\alpha}(t) = \frac{1}{J_t} [-m_p g l J_r \alpha(t) + m_p l r b_r \dot{\theta}(t) - J_p b_p \dot{\alpha}(t) - m_p r l \tau] \quad (1.7)$$

where:

$$J_t = J_p J_r - m_p^2 l^2 r^2 \quad (1.8)$$

Therefore, Eqs. 1.6 and 1.7 are useful to obtain the linear time invariant state space representation:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \end{aligned} \quad (1.9)$$

The system state $\mathbf{x}(t) \in \mathcal{R}^n$ defines the state variables needed to model the system $[\theta(t) \ \alpha(t) \ \dot{\theta}(t) \ \dot{\alpha}(t)]^T$. Here, $\theta(t)$ and $\dot{\theta}(t)$ are the angle position and angular velocity of

the arm, respectively, and $\alpha(t)$ and $\dot{\alpha}(t)$ are the angle position and angular velocity of the pendulum. The control input is $\mathbf{u}(t) \in \mathcal{R}^r$ stands for the motor voltage $v_m(t)$. Moreover, $\mathbf{y}(t) \in \mathcal{R}^m$ defines the outputs of the system considering the following state variables $[\theta(t) \ \alpha(t)]^T$ that are measured directly. Additionally, you must consider the rotatory pendulum parameters presented in Table 1.1

Remark 1: Table 1.1 contains the rotatory pendulum parameters of the Quanser Qube 3.



Table 1.1: Quanser Qube 3 pendulum system parameters.

Symbol	Parameter	Value
R_m	Terminal resistance	7.5 [Ω]
k_t	Torque constant	0.0422 [N m/A]
k_m	Motor back-emf constant	0.0422 [V/(rad/s)]
m_r	Rotary arm mass	0.095 [kg]
r	Rotary arm length (pivot to the end of the metal rod)	0.085 [m]
b_r	Rotary arm viscous damping	1.0×10^{-3} [N m/(rad/s)]
J_r	Rotary arm moment of inertia	2.2879×10^{-4} [kg m ²]
	$J_r = \frac{m_r r^2}{3}$	
m_p	Pendulum link mass	0.024 [kg]
L_p	Pendulum link length	0.129 [m]
b_p	Pendulum viscous damping	5.0×10^{-5} [N m/(rad/s)]
J_p	Pendulum moment of inertia	1.3313×10^{-4} [kg m ²]
	$J_p = \frac{m_p L_p^2}{3}$	
l	Half of the pendulum link length	0.0645 [m]
	$l = \frac{L_p}{2}$	

2. Development

D.1 In this Lab Practice, you must derive the linear state space model of the pendulum system using the Quanser Interactive Labs (QLabs) software. Then:

- Launch Matlab® and browse to the working directory that includes the Simulink models for this lab.
- Run the Quanser Interactive Labs (QLabs) software and login using your account.
- From the product list select *Qube 3 - Pendulum*, then select *Pendulum Workspace*. Hence, Quanser® Qube Servo 3 - Pendulum should look as Figure 1.1:

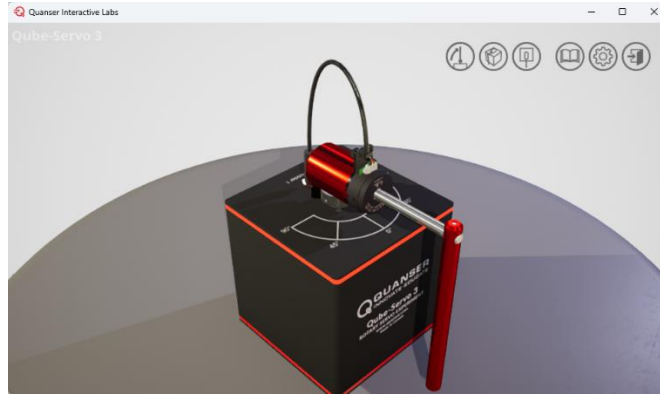


Fig. 1.1: Quanser® Quabe Servo 3 in the Rotatory Pendulum configuration.

D.2 Using Eqs. 1.6 and Equation 1.7, derive the linear state space model of the pendulum system, Eq. 1.9.

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t)\end{aligned}$$

D.3 Using the code *QuanserLabs_MR3003_Practice9_PartA_Script_CDSotelo.m*, follow the instructions below to obtain state space representation matrices derived in D.2:

- In “Step 1” of the code, load the model parameters defined in Table 1.1 and the total inertia J_t presented in Eq. 1.8.
- In “Step 2” of the code, program and compute the state space representation.

D.4 Build the Simulink® model shown in Fig. 1.2:

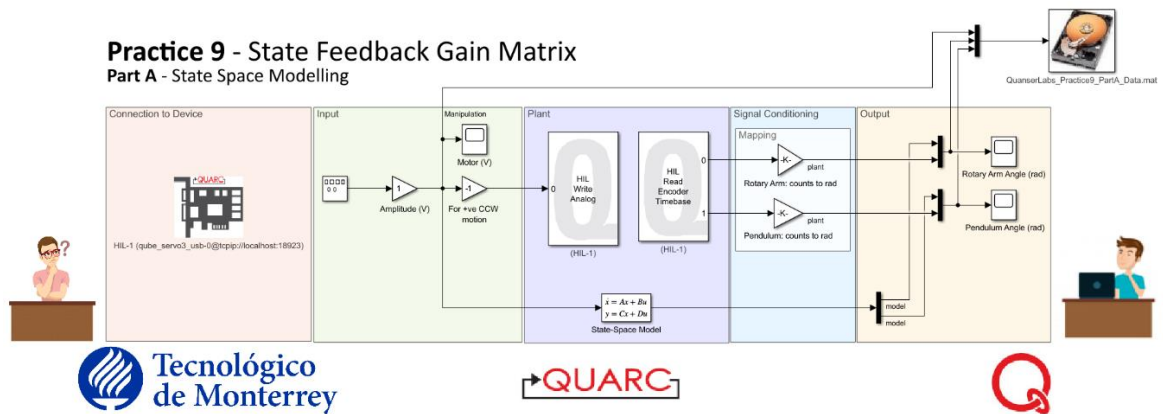


Fig. 1.2: Quanser® Quabe Servo 3 in the Rotatory Pendulum model validation.

Note: The output scopes display the responses of the rotary arm and pendulum angle of the Qube Servo 3 (green) in parallel with the response from the linear model of the system (dashed red).

Note: A square wave of 1 [V] with 1 [Hz] is applied to the pendulum system and state space mode.

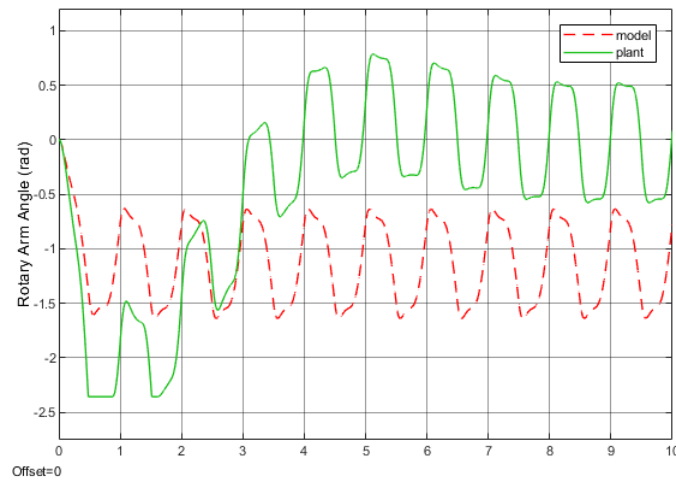
Note: Set the Amplitude Gain to 1 [V] to increase/decrease the amplitude of the wave.

Note: Add a Block Gain with value of -1 to guarantee that the servo (and thus the arm) turns in the CCW direction when the control voltage is positive $v_m(t) > 0$. It implies that in Mapping stage, the Rotary Arm: counts to rad block must be $-2 * \pi / (512 * 4)$.

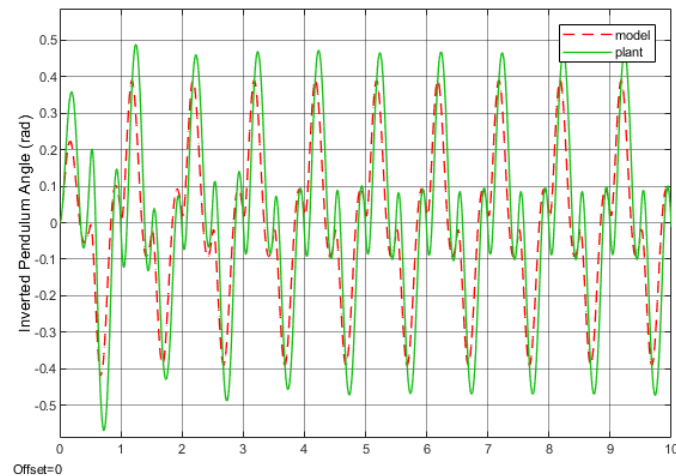
Then, follow the instructions below to validate the obtain state space representation matrices derived in D.2:

- i. Run for 10 sec and applies a step voltage input to the Qube Servo 3 and its state space model.
- ii. Build and run the QUARC controller.

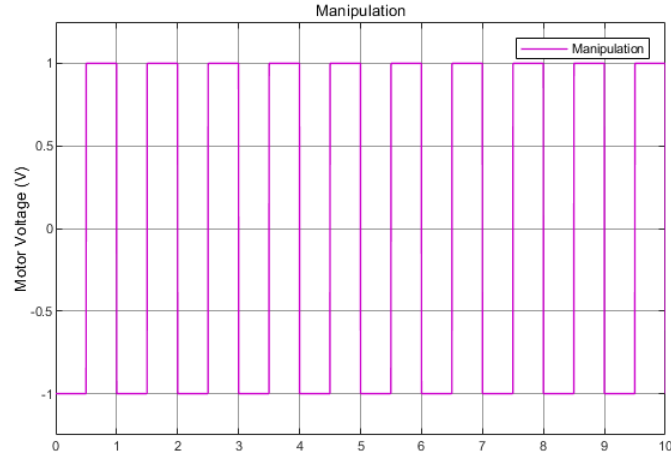
D.5 The scope response should be similar to Fig. 1.3. Attach the rotary pendulum response showing both the measured and simulated (model based) rotary arm and pendulum angles.



a)



b)



c)

Fig. 1.3: Step response of the rotary pendulum system.

D.6 In “Step 3” of *QuanserLabs_MR3003_Practice9_PartA_Script_CDSotelo.m*. Matlab code, generate plots for the rotary arm response $\theta(t)$ and pendulum response $\alpha(t)$ using the MATLAB script commands and the saved data in the *.MAT files, you must obtain a plot similar to Fig. 1.4.

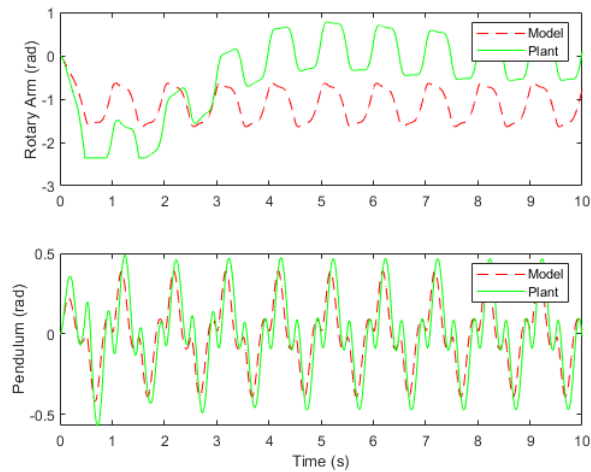


Fig. 1.4: Step response of the pendulum system.

D.6 Does your model represent the actual pendulum well? If not, explain why there might be discrepancies.

3. Report

R.1 According to D.2, derive the linear state space model of the pendulum system.

R.2 Based on D.3, compute the state space representation matrices derived in D.2.

***Note:** Add your Matlab® code used to compute these values.

R.3 Based on the Virtual Quanser® Qube Servo 3 - Pendulum in D.1 and your state space model, obtain the scopes shown in Fig. 1.3, which represents the response of the rotary pendulum system for the virtual lab and the model and.

***Note:** Add your Matlab® code used to compute these values.

R.4 Considering D.6, generate plots for the rotary arm response $\theta(t)$ and pendulum response $\alpha(t)$ using the MATLAB script commands and the saved data in the *.MAT files.

***Note:** Add your Matlab® code used to compute this value.

R.5 Conclusions.